Hackathon Technical Solution Report

Field	Details
Problem ID	P1
Problem Title	Robust Low-Light Image Restoration and Enhancement
Team Name	VisionForge
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GitHub Repository	https://github.com/Vaishnavi-Dhanuka/CV-Hackathon-Project
Live Demo / Deployment	https://lumina-streamlit.onrender.com

1. Problem Understanding & Solution Strategy

Low-light images tend to suffer from low visibility, noise amplification, colour distortion, and lack of detail. LUMINA offers a user-friendly interface for comparing classic computer vision enhancement, RetinexFormer trained model and experimental hybrid approach.

The tool utilizes image enhancement techniques that seek to increase visibility without losing details and natural colours. As per the documented assessment, the use of pretrained RetinexFormer alone is advised; the existing hybrid approach is still experimental.

2. Technical Architecture & Implementation

2.1 Classic Algorithms Baseline

The classic way implies illumination preprocessing with Retinex methods and guided filtering through the project's classic adapter. Image processing is done with the help of `hybrid/classical_client.py` .

2.2 Deep Learning Approach

Deep learning way assumes the use of pre-trained RetinexFormer model. Processing and returning an enhanced image is done through the FastAPI `/enhance` endpoint. Pre-trained weights are not saved in the Git repository.

2.3 Hybrid Approach & Adaptivity Mechanism

The hybrid way combines classic and RetinexFormer outputs using spatially-adaptive heuristic weights. Weights regulate the influence of each of outputs at each region of the image. Weights are not attention maps but blending proportions. According to the evaluation results, the proposed fusion experiment did not beat the RetinexFormer method alone.

3. Experimental Benchmarks & Evaluation

3.1 Quantitative Benchmarks

The below results are from the 15-image held-out LOL eval15 comparison. Both LPIPS and NIQE have been put in the perceptual metric column. The latency was not calculated for this comparison.

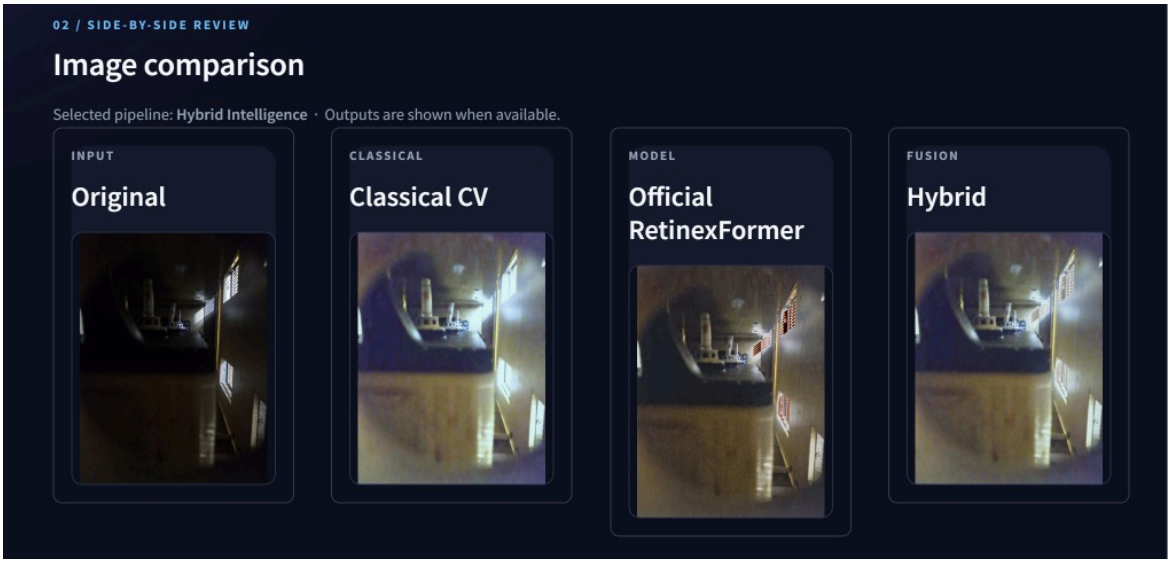
Approach / Model Variant	PSNR (dB)	SSIM	LPIPS / NIQE	Latency
Classical Baseline	16.3686	0.5048	0.3590 / 7.9804	Not recorded
Deep Learning Only (RetinexFormer)	21.7720	0.8224	0.1524 / 3.8765	Not recorded
Proposed Hybrid System	18.1390	0.5885	0.2943 / 7.0345	Not recorded

RetinexFormer outperformed the other approaches in terms of average performance on the listed quality metrics. The combined approach produced improvements on some of the images; however, no improvements were found on the average measures. Therefore, the standalone approach is preferable.

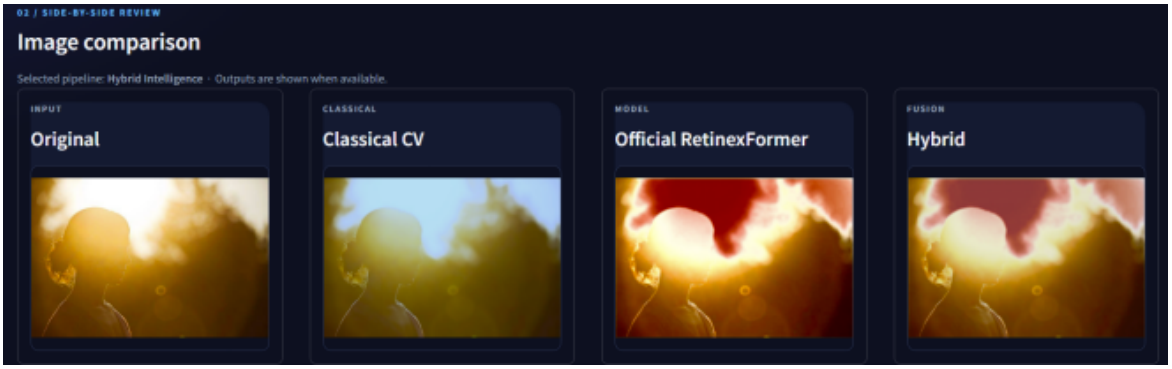
3.2 Visual Outputs & Intermediate Representations

The screenshots below show two real LUMINA side-by-side comparisons and an illumination map generated by the hybrid pipeline.

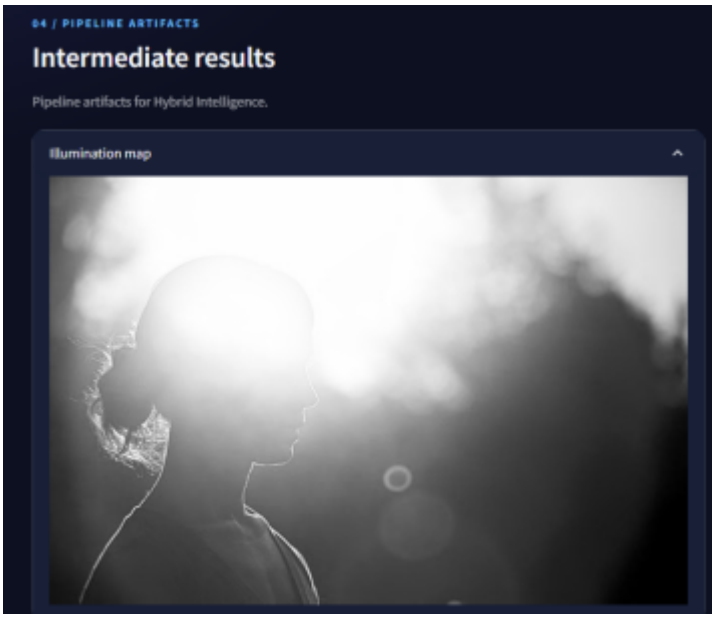
Visual output 1 — Low-light indoor scene: original, classical, RetinexFormer and hybrid



Visual output 2 — Backlit scene comparison



Intermediate representation — Illumination map



3.3 Ablation Study

The comparison involved 50:50 fixed fusion, adaptive fusion, spatial adaptive fusion, and RetinexFormer-dominant fusion. All four fusion techniques did not satisfy the criteria required to be selected as a substitute for RetinexFormer on its own. It means that the present fusion method requires further fine-tuning to be considered better.

4. Failure Modes & Limitations

The hybrid technique has underperformed RetinexFormer individually for the mentioned average performance metrics. Classic image enhancement techniques have the ability to create color shifts or over-brighten images in hard conditions, whereas strong enhancement might introduce noise in darker parts.

The weights used by the hybrid model are heuristics that are not able to determine the correctness of image enhancement. Better weights for noisy images, highlight preservation, and color correction might improve the situation.

5. Deployment & Quickstart Guide

Quickstart Commands

Initialize the repository submodule:

```
git submodule update --init --recursive
```

Start the backend API:

```
python -m uvicorn deep_learning.api:app --host 127.0.0.1 --port 8000
```

In a second terminal, start the Streamlit frontend:

```
python -m streamlit run app/main.py
```

Install the project dependencies with `pip install -r requirements.txt` and provision the separately distributed RetinexFormer checkpoint as described in `LUMINA_SETUP_GUIDE.md`.

Deployment Format

LUMINA employs Streamlit as its frontend and FastAPI as its inference backend. The frontend is deployed at the above-mentioned public URL. The inference backend, which was running on Render Free, exceeded the memory limit of 512 MB while performing inference. Therefore, hosted inference using the model is unconfirmed.

6. AI Assistance Declaration

AI assistance was used to support system design; the team reviewed the suggestions and is responsible for the final implementation and report. The team acknowledges the hackathon policy that AI-generated content in the final report must not exceed **30%**, with proportional report-mark penalties for exceeding the limit.